

Manual

Graphic Display of Energy Counters EMG-GEL 8.3

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1 Overview: Graphic Display of Energy Counters

Purpose

The graphic display of energy counters is based on the Graphic Machinery and visualizes a specific level of a company's machinery (production hall, foreman's area,...). You can add energy counter resources to this graphic machinery. The display does not only integrate master data, but also current actual data collected as part of the energy data collection.

Implementation notes

You use the function package for the following purposes:

- You want to display the available energy counters of your company and current actual data.

Integration

The application uses the current EMG data (e.g. counter status, current consumption) for the display. The energy monitor provides this data in a table (EMG-GEM).

Features

- Graphic display
 - Graphic display of a company's machinery with energy counters: production hall, department, supervisor's area, etc.
- Display of the current counter status
 - Display of the current resource status of the counter resources using different colors
- Tooltip
 - The info window (tooltip) provides detailed information
- Graphic editor
 - Graphic editor to create layouts of halls where the symbols for energy counters, machines, machine groups and production lines can be placed according to your requirements.

2 Graphic Machinery

Summary

Menu	Production facilities management → Current information → Graphic machinery Information management → Production overview → Graphic machinery
Transaction code	mpark
Function authorization	mpark

Usage

Graphic machinery provides a very striking presentation of essential workplace and production information shown in terms of time and space. It allows various presentations (layouts) of a random selection of organizational units, which the user can create as required.

On the one hand, graphic machinery can serve as an indicator for the production manager, and on the other hand, it can be shown on large screens in the production department as a continuous overview of production status.

The user interface

The graphic machinery user interface differs depending on the mode the user is in. Here, we differentiate between the following modes:

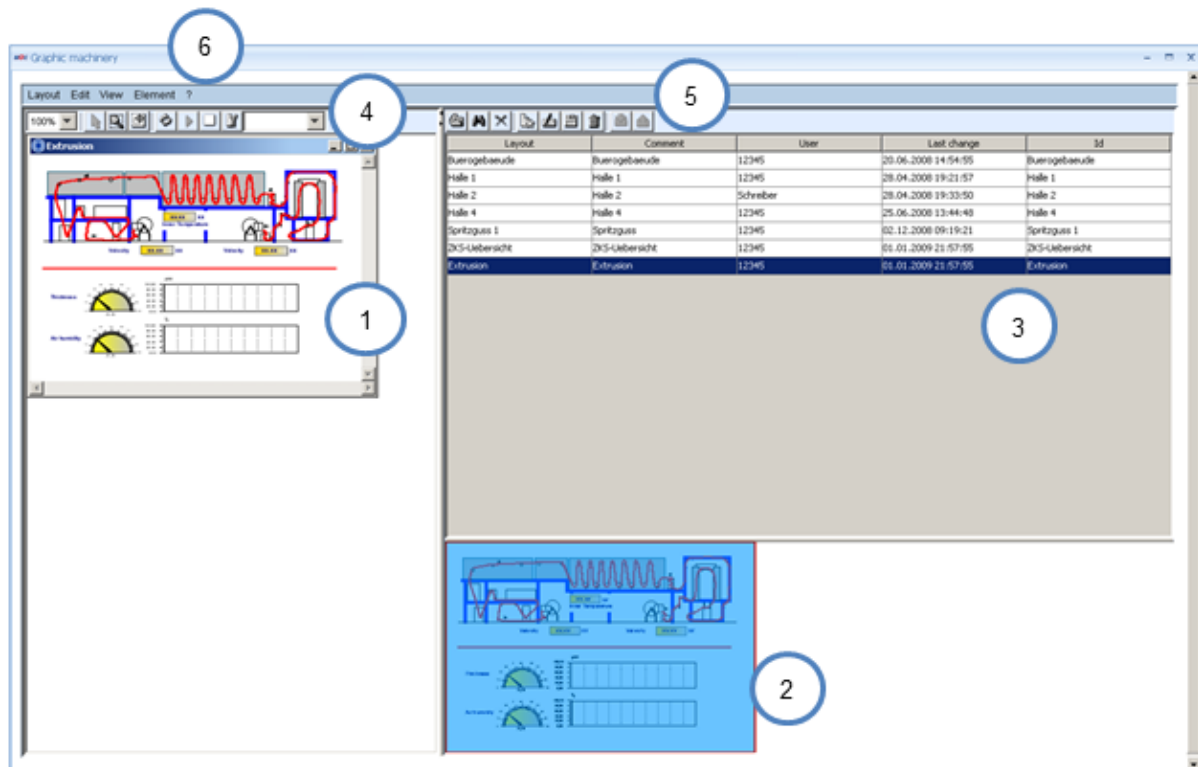
- [View mode](#) → this is where current data are displayed, where already existing layouts can be selected or new ones can be created.
- [Edit mode](#) → this is where, among other things, objects that have defined templates can be assigned to a layout so that it can be displayed in view mode.

3 View mode

In view mode, among other things, the layouts that were created are displayed with their individual elements and current data. In addition, display options can be selected and new layouts created.

Sections

We differentiate among the following sections of the user interface:



1 Active, open layout window

The selected layouts and their component objects are displayed in this section.

2 Overview panel

The overview panel provides an overview of the current open layout and shows the user which part of the layout is currently displayed in the layout window.

3 Table of available layouts

To be able to navigate between these layouts, there is a table on the right side of the screen that contains the names of the separate layouts and information about them. Using this table, the user can switch to the different layouts.

By double-clicking the left mouse button on a line of the table, the display switches to that particular layout. The name of the active layout is shown in the field at the upper edge of the window that opens.

4 Viewer toolbar

You can select display options from the viewer toolbar and switch to edit mode.

5 Explorer toolbar

Layouts are created and maintained in the explorer toolbar.

6 Menu bar

Explorer and viewer toolbar functions, among other things, can be accessed from the menu bar.

Viewer toolbar



Combo box "zoom factor"

Predefined zoom factors can be selected, or user-defined zoom factors entered here.

The "Select" button

By clicking on this button, the mouse is set to "select" mode. In this mode detail views can be opened with a double-click. By right-clicking on a HYDRA object, the context menu opens.

The "Zoom rectangle" button

By clicking on this button, the mouse is set to "zoom rectangle" mode. In this mode, you can draw a rectangle around the layout, to which the layout is then sized.

The "Full screen view" button

Here, the active layout is zoomed in to full-screen size. You exit full-screen view by clicking the mouse or pressing space bar, return or escape.

The "Update all layouts" button

This button updates all currently opened layouts.

The "Start update" button

This starts the automatic update. Current values are now transferred to the layouts from the data server and they are updated accordingly.

The "Stop update" button

This stops the automatic update.

The "Switch to editor mode" button

Clicking on this button switches machinery to editor mode.

Combo box "Layout arrangement"

Using this combo box, you can determine how the currently open layouts should be arranged.

Explorer toolbar



The "Load layout" button

This button opens the layout window for the layout entry selected in the table.

The "Search for layout" button

Clicking on this button opens the "search" dialog.

Here, you can define the object type (machine, line, terminal, ...) and attribute (number, designation, ...) you would like to search for, and then type the search value in the text field. After clicking on the search button, the dialog closes and the layouts containing the search object are displayed and shown highlighted in the explorer table. If a highlighted layout is open, the objects found in it are marked with handles.

The "Delete search" button

Clicking on this button removes the highlighting for the layouts found in the explorer table. The search criteria are removed from the session.

The "New layout" button

Clicking this button opens a new layout window. The respective entry is incorporated into the table.

The "Rename layout" button

After clicking on this button, the "Rename layout" dialog appears.

Here, you can edit the layout name as well as the associated comment.

The "Save layout" button

Clicking on this button saves the layout selected in the table to the server.

The "Delete layout" button

Clicking on this button removes the selected layout from the table and closes the associated layout window.

The "Import layout" button

This function is currently not available (future use).

The "Export layout" button

This function is currently not available (future use).

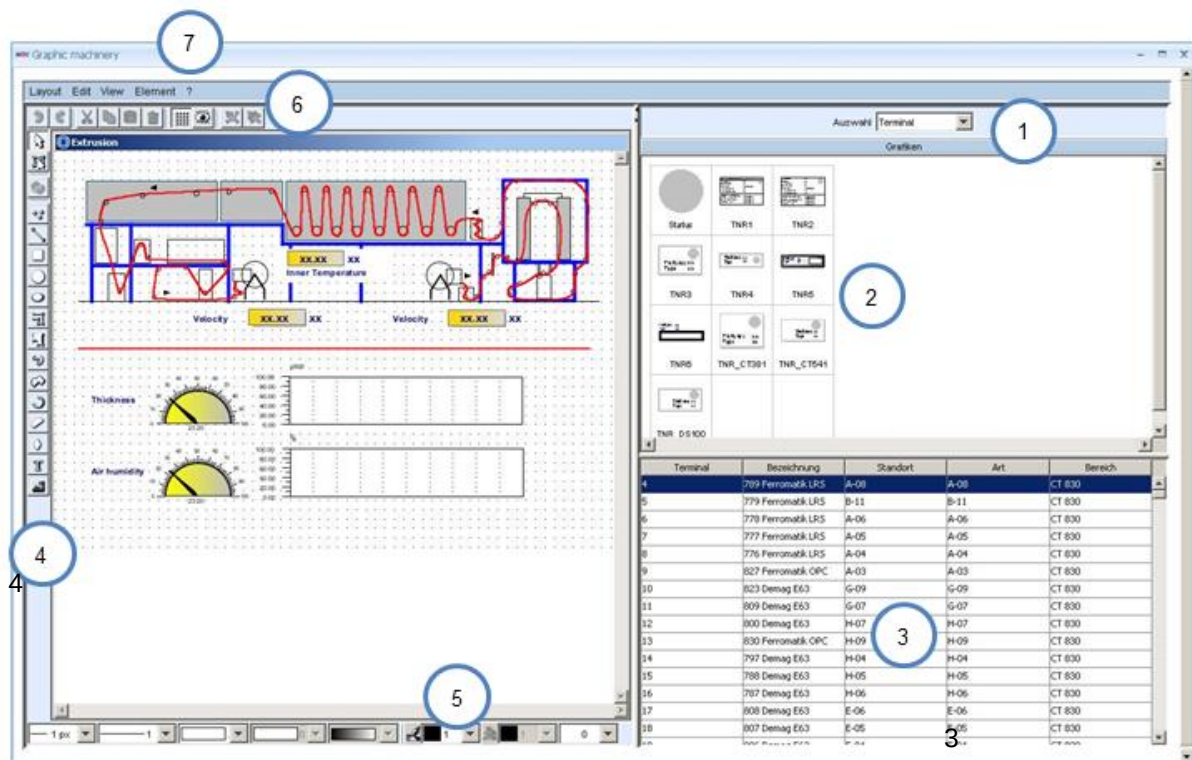
4 Edit mode

You can integrate individual objects using a presentation template and other configurations into each layout, and you can also add standard geometries to them in edit mode. This is available in view mode by clicking on the "Switch to edit mode" button.

By integrating a defined object, e.g. a machine, using a presentation template, it is displayed in view mode with the information currently defined in the presentation template like the status, for example.

Sections

The user interface is divided into the following sections:



Select object type

The following object types are currently available for selection (the relevant HYDRA products are entered in parentheses):

- Machine (BDE/MDE)
- Machine group (BDE/MDE)
- Terminal (MW)
- Material buffer (MPL)
- Lines (MDE)
- Accesses (ZKS)
- Access groups (ZKS)
- Resources (WRM/EMG; resources that are no DNC resources and not configured as "file-based")

2 Preview of the presentation template

In this section you can see a preview of the presentation template available for an object.

3 Available objects

The objects available to an object type in the system(s) connected respectively are listed in a table in the available objects section.

4 Toolbar with standard geometries

The standard geometries available here can also be integrated into layouts.

5 Toolbar with graphic styles

Standard geometries can be configured from the toolbar with graphic styles.

6 Editor toolbar

Among other things, the editor tool bar can be used to configure the display and to switch to view mode.

7 Menu bar

The menu bar can be used, among other things, to call up editor toolbar functions and to switch to view mode.

Editor toolbar**The "Undo" button**

By clicking on this button the last action performed is undone.

The "Redo" button

By clicking on this button the last action undone is restored.

The "Cut" button

By clicking on this button, the elements selected are copied to the clipboard. When the elements are pasted, the original elements are removed.

The "Copy" button

When you click on this button, the elements selected are copied to the clipboard.

The "Paste" button

When you click on this button, the elements on the clipboard are copied into the layout.

The "Delete" button

Clicking on this button deletes the selected elements.

The "Grid on/off" button

The drawing grid can be activated/ deactivated by clicking on this button.

The "View mode" button

Clicking on this button switches machinery to view mode.

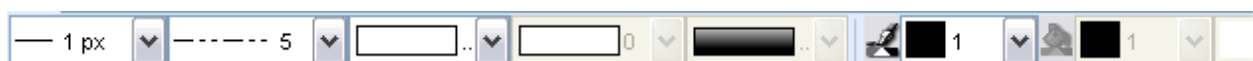
The "Create group" button

Clicking on this button compiles the selected elements into a group.

The "Cancel group" button

If a group is selected in the layout, it can be canceled by clicking on this button.

Graphic styles toolbar



Combo box "Line thickness"

The line thickness of the selected element can be defined here.

Combo box "Line type"

The line thickness of the selected element can be defined here.

Combo box "Fill type"

The fill type can be defined here. Normal color, pattern and color gradient are the choices.

Combo box "Fill pattern"

If "Pattern" is selected as fill type, a pattern may be selected here.

Combo box "Color gradient"

If "Color gradient" is selected as fill type, the type of gradient may be defined here.

Combo box "Line color"

The line color of the selected elements can be defined here.

Combo box "Fill color 1"

This combo box is used if "normal color" was selected as fill type. The fill color is selected here.

Combo box "Fill color 2"

If "Color gradient" is selected as fill type, the second color may be selected here.

Standard geometry toolbar



The "Select" button

Sets the mouse mode to "Select". Elements can be selected, moved and scaled.

The "Select in group" button

If there is a group in the layout, an element within a group can be selected using this mouse mode.

The "Rotate" button

The mouse mode "Rotate" is set here. A green dot appears on the selected element that represents its center of rotation. The center of rotation can be moved. The element can be moved around the center of rotation by holding down the left mouse button.

The "Points" button

Used to set points in the layout.

The "Line" button

Used to draw lines in the layout.

The "Rectangle" button

Used to draw a rectangle.

The "Circle" button

Used to draw a circle.

The "Ellipse" button

Used to draw an ellipse.

The "Polyline" button

Used to draw a polyline.

The "Polygon" button

Used to draw a polygon.

The "Open spline" button

Used to draw an open spline.

The "Closed spline" button

Used to draw a closed spline.

The "Arc" button

Used to draw an arc.

The "Circle segment" button

Used to draw a segment of a circle.

The "Sector" button

Used to draw a sector of a circle.

The "Insert text" button

Fixed texts can be placed in the layout using this button.

"Insert BMP graphic" button

Existing images can be inserted into the layout using this button.

Properties dialogs in the editor

The properties dialog of a graphic element is opened in the editor by "right click-->properties" or double-click. If this is a standard geometry, the geometric properties of the element can be modified using the first tab. The "Line style" and "Fill style" tabs contain options that allow you to manipulate the graphic style of the element.

The properties of the object are displayed in the properties dialog of a HYDRA object. In addition, the templates are assigned here.

The presentation template is the template with which the HYDRA object is displayed.

You open the detail view template in the viewer by double-clicking on the object.

The context menu template determines the appearance and behavior of this object's context menu.

The tool tip template is used as tool tip for this object.

Edit layout

Integrate an object

After selecting an object type, e.g. machine, terminal, all the presentation templates and objects available for this object type are displayed in the preview.

After you have chosen the object you would like to select (highlighted in the table) and have then selected a presentation template in which the object is displayed in the corresponding layout, single click on the open layout to insert the object and display it using the selected presentation template.

By switching to view mode, the current values of the inserted objects are then received from the data server and updated accordingly.

A defined object can be integrated into a layout more than once by selecting it several times from the objects list.

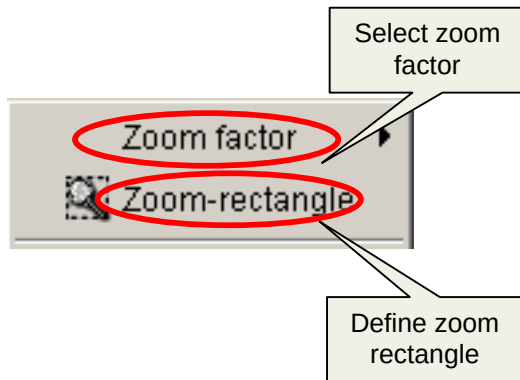
Select objects

Individual objects of a layout can be selected by clicking on them, or via a window using the mouse, and they can then be edited.

If you want to highlight multiple objects, you must either use a window or click on the individual objects while holding down the "shift" key.

Zoom in edit mode

You can use the zoom function in edit mode by clicking on the "View" menu item. Here, you can select a zoom factor or use the zoom rectangle to select an image section to enlarge it, so that it is displayed filling out the layout window completely.



5 Shortcuts

Menu item "Layout"

Open	Ctrl. + O
New	Ctrl. + N
Rename layout	Ctrl. + R
Save	Ctrl. + S
Delete layout	Ctrl. + D
Import layout (future use)	Ctrl. + I
Export layout (future use)	Ctrl. + E

Menu item "Edit"

Undo	Ctrl. + Z
Restore	Ctrl. + Y
Cut	Ctrl. + X
Copy	Ctrl. + C
Paste	Ctrl. + V
Delete	Del
Select all	Ctrl. + A

Menu item "Element"

In the background	Ctrl. + B
In the foreground	Ctrl. + F
Create icon	Ctrl. + G
Cancel icon	Ctrl. + U
Properties	ALT + Enter

Menu item "Update"

Update	F5
Begin update	F7
Stop update	F9